



National  
Qualifications  
2015

**X757/75/02**

**Physics**  
**Section 1—Questions**

TUESDAY, 5 MAY

9:00 AM – 11:00 AM

Instructions for the completion of Section 1 are given on *Page two* of your question and answer booklet X757/75/01.

Record your answers on the answer grid on *Page three* of your question and answer booklet.

Reference may be made to the Data Sheet on *Page two* of this booklet and to the Relationship Sheet X757/75/11.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



\* X 7 5 7 7 5 0 2 \*

## DATA SHEET

### Speed of light in materials

| Material       | Speed in $\text{m s}^{-1}$ |
|----------------|----------------------------|
| Air            | $3.0 \times 10^8$          |
| Carbon dioxide | $3.0 \times 10^8$          |
| Diamond        | $1.2 \times 10^8$          |
| Glass          | $2.0 \times 10^8$          |
| Glycerol       | $2.1 \times 10^8$          |
| Water          | $2.3 \times 10^8$          |

### Gravitational field strengths

|         | Gravitational field strength on the surface in $\text{N kg}^{-1}$ |
|---------|-------------------------------------------------------------------|
| Earth   | 9.8                                                               |
| Jupiter | 23                                                                |
| Mars    | 3.7                                                               |
| Mercury | 3.7                                                               |
| Moon    | 1.6                                                               |
| Neptune | 11                                                                |
| Saturn  | 9.0                                                               |
| Sun     | 270                                                               |
| Uranus  | 8.7                                                               |
| Venus   | 8.9                                                               |

### Specific latent heat of fusion of materials

| Material       | Specific latent heat of fusion in $\text{J kg}^{-1}$ |
|----------------|------------------------------------------------------|
| Alcohol        | $0.99 \times 10^5$                                   |
| Aluminium      | $3.95 \times 10^5$                                   |
| Carbon Dioxide | $1.80 \times 10^5$                                   |
| Copper         | $2.05 \times 10^5$                                   |
| Iron           | $2.67 \times 10^5$                                   |
| Lead           | $0.25 \times 10^5$                                   |
| Water          | $3.34 \times 10^5$                                   |

### Specific latent heat of vaporisation of materials

| Material       | Specific latent heat of vaporisation in $\text{J kg}^{-1}$ |
|----------------|------------------------------------------------------------|
| Alcohol        | $11.2 \times 10^5$                                         |
| Carbon Dioxide | $3.77 \times 10^5$                                         |
| Glycerol       | $8.30 \times 10^5$                                         |
| Turpentine     | $2.90 \times 10^5$                                         |
| Water          | $22.6 \times 10^5$                                         |

### Speed of sound in materials

| Material       | Speed in $\text{m s}^{-1}$ |
|----------------|----------------------------|
| Aluminium      | 5200                       |
| Air            | 340                        |
| Bone           | 4100                       |
| Carbon dioxide | 270                        |
| Glycerol       | 1900                       |
| Muscle         | 1600                       |
| Steel          | 5200                       |
| Tissue         | 1500                       |
| Water          | 1500                       |

### Specific heat capacity of materials

| Material  | Specific heat capacity in $\text{J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$ |
|-----------|---------------------------------------------------------------------------|
| Alcohol   | 2350                                                                      |
| Aluminium | 902                                                                       |
| Copper    | 386                                                                       |
| Glass     | 500                                                                       |
| Ice       | 2100                                                                      |
| Iron      | 480                                                                       |
| Lead      | 128                                                                       |
| Oil       | 2130                                                                      |
| Water     | 4180                                                                      |

### Melting and boiling points of materials

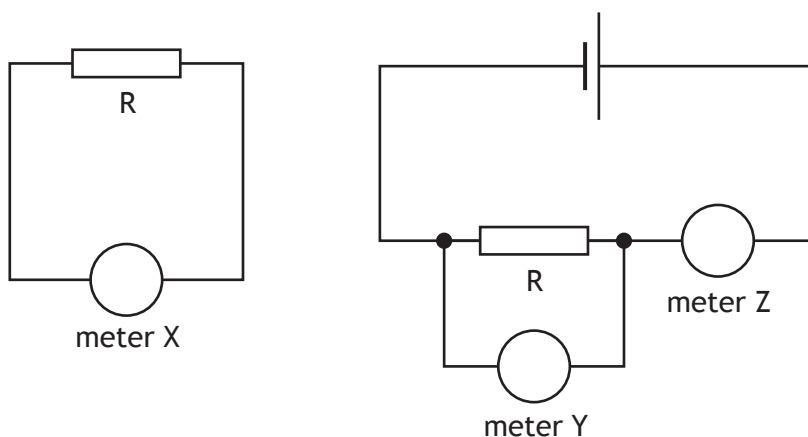
| Material  | Melting point in $^\circ\text{C}$ | Boiling point in $^\circ\text{C}$ |
|-----------|-----------------------------------|-----------------------------------|
| Alcohol   | -98                               | 65                                |
| Aluminium | 660                               | 2470                              |
| Copper    | 1077                              | 2567                              |
| Glycerol  | 18                                | 290                               |
| Lead      | 328                               | 1737                              |
| Iron      | 1537                              | 2737                              |

### Radiation weighting factors

| Type of radiation | Radiation weighting factor |
|-------------------|----------------------------|
| alpha             | 20                         |
| beta              | 1                          |
| fast neutrons     | 10                         |
| gamma             | 1                          |
| slow neutrons     | 3                          |
| X-rays            | 1                          |

**SECTION 1**  
**Attempt ALL questions**

1. Two circuits are set up as shown.



Both circuits are used to determine the resistance of resistor  $R$ .

Which row in the table identifies meter  $X$ , meter  $Y$  and meter  $Z$ ?

|   | <i>meter X</i> | <i>meter Y</i> | <i>meter Z</i> |
|---|----------------|----------------|----------------|
| A | ohmmeter       | voltmeter      | ammeter        |
| B | ohmmeter       | ammeter        | voltmeter      |
| C | voltmeter      | ammeter        | ohmmeter       |
| D | ammeter        | voltmeter      | ohmmeter       |
| E | voltmeter      | ohmmeter       | ammeter        |

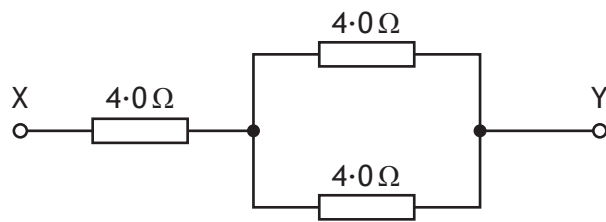
2. Which of the following statements is/are correct?

- I The voltage of a battery is the number of joules of energy it gives to each coulomb of charge.
- II A battery only has a voltage when it is connected in a complete circuit.
- III Electrons are free to move within an insulator.

- A I only
- B II only
- C III only
- D II and III only
- E I, II and III

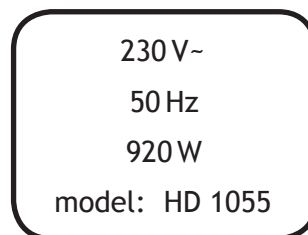
[Turn over

3. A circuit is set up as shown.



The resistance between X and Y is

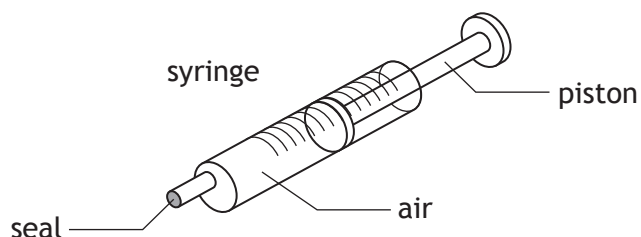
- A  $1.3\ \Omega$
  - B  $4.5\ \Omega$
  - C  $6.0\ \Omega$
  - D  $8.0\ \Omega$
  - E  $12\ \Omega$ .
4. The rating plate on an electrical appliance is shown.



The resistance of this appliance is

- A  $0.017\ \Omega$
- B  $0.25\ \Omega$
- C  $4.0\ \Omega$
- D  $18.4\ \Omega$
- E  $57.5\ \Omega$ .

5. A syringe containing air is sealed at one end as shown.



The piston is pushed in slowly.

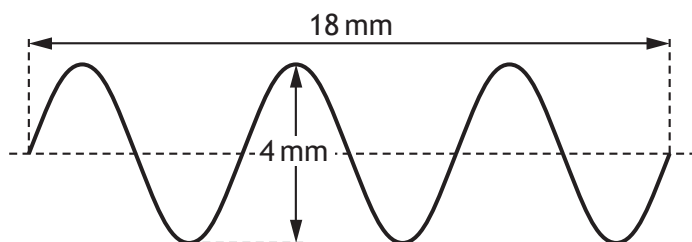
There is no change in temperature of the air inside the syringe.

Which of the following statements describes and explains the change in pressure of the air in the syringe?

- A The pressure increases because the air particles have more kinetic energy.
  - B The pressure increases because the air particles hit the sides of the syringe more frequently.
  - C The pressure increases because the air particles hit the sides of the syringe less frequently.
  - D The pressure decreases because the air particles hit the sides of the syringe with less force.
  - E The pressure decreases because the air particles have less kinetic energy.
6. The pressure of a fixed mass of gas is 150 kPa at a temperature of 27 °C.  
The temperature of the gas is now increased to 47 °C.  
The volume of the gas remains constant.  
The pressure of the gas is now
- A 86 kPa
  - B 141 kPa
  - C 150 kPa
  - D 160 kPa
  - E 261 kPa.

[Turn over

7. The diagram represents a water wave.



The wavelength of the water wave is

- A 2 mm
  - B 3 mm
  - C 4 mm
  - D 6 mm
  - E 18 mm.
8. A student makes the following statements about different types of electromagnetic waves.
- I Light waves are transverse waves.
  - II Radio waves travel at  $340 \text{ m s}^{-1}$  through air.
  - III Ultraviolet waves have a longer wavelength than infrared waves.

Which of these statements is/are correct?

- A I only
  - B I and II only
  - C I and III only
  - D II and III only
  - E I, II and III
9. Alpha radiation ionises an atom.
- Which statement describes what happens to the atom?
- A The atom splits in half.
  - B The atom releases a neutron.
  - C The atom becomes positively charged.
  - D The atom gives out gamma radiation.
  - E The atom releases heat.

10. A sample of tissue is irradiated using a radioactive source.

A student makes the following statements.

The equivalent dose received by the tissue is

- I reduced by shielding the tissue with a lead screen
- II increased as the distance from the source to the tissue is increased
- III increased by increasing the time of exposure of the tissue to the radiation.

Which of the statements is/are correct?

- A I only
  - B II only
  - C I and II only
  - D II and III only
  - E I and III only
11. A sample of tissue receives an absorbed dose of  $16\text{ }\mu\text{Gy}$  from alpha particles.  
The radiation weighting factor for alpha particles is 20.  
The equivalent dose received by the sample is
- A  $0.80\text{ }\mu\text{Sv}$
  - B  $1.25\text{ }\mu\text{Sv}$
  - C  $4\text{ }\mu\text{Sv}$
  - D  $36\text{ }\mu\text{Sv}$
  - E  $320\text{ }\mu\text{Sv}$ .
12. For a particular radioactive source, 240 atoms decay in 1 minute.  
The activity of this source is
- A 4 Bq
  - B 180 Bq
  - C 240 Bq
  - D 300 Bq
  - E 14 400 Bq.

[Turn over

13. The letters X, Y and Z represent missing words from the following passage.

*During a nuclear .....X..... reaction two nuclei of smaller mass number combine to produce a nucleus of larger mass number. During a nuclear .....Y..... reaction a nucleus of larger mass number splits into two nuclei of smaller mass number. Both of these reactions are important because these processes can release .....Z..... .*

Which row in the table shows the missing words?

|   | X       | Y       | Z         |
|---|---------|---------|-----------|
| A | fusion  | fission | electrons |
| B | fission | fusion  | energy    |
| C | fusion  | fission | protons   |
| D | fission | fusion  | protons   |
| E | fusion  | fission | energy    |

14. Which of the following quantities is fully described by its magnitude?

- A Force
- B Displacement
- C Energy
- D Velocity
- E Acceleration

15. The table shows the velocities of three objects X, Y and Z over a period of 3 seconds. Each object is moving in a straight line.

|                                     |   |   |   |   |
|-------------------------------------|---|---|---|---|
| Time (s)                            | 0 | 1 | 2 | 3 |
| Velocity of X ( $\text{m s}^{-1}$ ) | 2 | 4 | 6 | 8 |
| Velocity of Y ( $\text{m s}^{-1}$ ) | 0 | 1 | 2 | 3 |
| Velocity of Z ( $\text{m s}^{-1}$ ) | 0 | 2 | 5 | 9 |

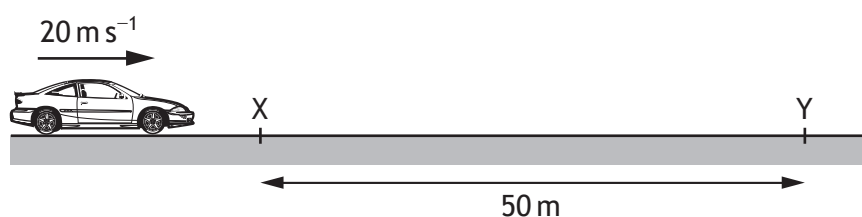
Which of the following statements is/are correct?

- I X moves with constant velocity.
- II Y moves with constant acceleration.
- III Z moves with constant acceleration.

- A I only
- B II only
- C I and II only
- D I and III only
- E II and III only

16. A car of mass 1200 kg is travelling along a straight level road at a constant speed of  $20 \text{ m s}^{-1}$ .

The driving force on the car is 2500 N. The frictional force on the car is 2500 N.



The work done moving the car between point X and point Y is

- A 0 J
- B 11 800 J
- C 125 000 J
- D 240 000 J
- E 250 000 J.

[Turn over

17. A person sits on a chair which rests on the Earth. The person exerts a downward force on the chair.



Which of the following is the reaction to this force?

- A The force of the chair on the person
  - B The force of the person on the chair
  - C The force of the Earth on the person
  - D The force of the chair on the Earth
  - E The force of the person on the Earth
18. A package falls vertically from a helicopter. After some time the package reaches its terminal velocity.

A group of students make the following statements about the package when it reaches its terminal velocity.

- I The weight of the package is less than the air resistance acting on the package.
- II The forces acting on the package are balanced.
- III The package is accelerating towards the ground at  $9.8 \text{ m s}^{-2}$ .

Which of these statements is/are correct?

- A I only
- B II only
- C III only
- D I and III only
- E II and III only

19. The distance from the Sun to Proxima Centauri is 4.3 light years.

This distance is equivalent to

- A  $1.4 \times 10^8 \text{ m}$
- B  $1.6 \times 10^{14} \text{ m}$
- C  $6.8 \times 10^{14} \text{ m}$
- D  $9.5 \times 10^{15} \text{ m}$
- E  $4.1 \times 10^{16} \text{ m}$ .

20. Light from a star is split into a line spectrum of different colours. The line spectrum from the star is shown, along with the line spectra of the elements calcium, helium, hydrogen and sodium.



The elements present in this star are

- A sodium and calcium
- B calcium and helium
- C hydrogen and sodium
- D helium and hydrogen
- E calcium, sodium and hydrogen.

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2  
OF YOUR QUESTION AND ANSWER BOOKLET]

## ACKNOWLEDGEMENTS

Question 17 – Rob Byron/shutterstock.com